

LEVEL 6 · CHALLENGE

The Last Door

Everything the lock needs is in `core.sealed` and `carrier.wav`. Your job is to find the five numbers below — each a small whole number, the answer to one problem. Earn all five and the core opens.

Use these constants in every calculation, and round each final answer to the nearest whole unit asked for:

G	$6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$
c	$2.998 \times 10^8 \text{ m/s}$
Mass of the Sun	$1.989 \times 10^{30} \text{ kg}$

Door I — The Collapsed Star (key 1)

A black hole of ten solar masses. There is a radius from which not even light escapes — the one Schwarzschild found in the field equations. Key 1 = that radius, in whole metres.

Door II — The Cloud That Would Collapse (key 2)

A cold cloud of gas: 20 K, 10^3 particles per cm^3 , mean molecular weight 2.3. There is a mass above which such a cloud can no longer hold itself up and begins to fall together into stars — the threshold Jeans wrote down. Key 2 = that mass, in whole solar masses.

Door III — The Fleeing Galaxy (key 3)

A galaxy's light comes in stretched — redshift $z = 0.030$. The faster a galaxy flees, the farther it lies, by the law Hubble found (take $H_0 = 70 \text{ km/s per Mpc}$). How far away is it? Key 3 = its distance, in whole megaparsecs.

Door IV — The Slow Orbit (key 4)

A small body circles a Sun-like star at a distance of 4.0 AU. The same law of motion you can derive from least action fixes how long one lap takes. Key 4 = its orbital period, in whole days.

Door V — The Carrier (key 5)

Open `carrier.wav`. Under the hiss, one note rings louder than the rest. Find the frequency of that strongest note. Key 5 = that frequency, in whole hertz. (If the number rings a bell for a radio astronomer, you're on the right track.)

A hint, for the curious: notice how each door is a light, quick thing set against a heavy, slow one — a pull, a background, a carrier. Separate the fast from the slow and each problem comes apart in your hands.

Files	core.sealed, carrier.wav
Needs	Python 3 (core.sealed explains what the lock expects)
Flag format	not disclosed — you'll know it when the core opens